



# NORTH SYDNEY BOYS HIGH SCHOOL

## 2013 Year 11 Task 2

# Mathematics

## Extension 1

### General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

### Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Fletcher
- ☐ Mr Ireland
- ☐ Mr Lowe
- ☐ Mr Lucas
- ☐ Mrs Ziazaris

Student Name: \_\_\_\_\_

(To be used by the exam markers only.)

| Question No | 1-6           | 7             | 8             | 9             | 10            | 11            | 12            | Total           | Total %           |
|-------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|-----------------|-------------------|
| Mark        | $\frac{6}{9}$ | $\frac{9}{9}$ | $\frac{9}{9}$ | $\frac{9}{9}$ | $\frac{9}{9}$ | $\frac{9}{9}$ | $\frac{9}{9}$ | $\frac{60}{60}$ | $\frac{100}{100}$ |



Student Name : \_\_\_\_\_

**SECTION 1:        MULTIPLE CHOICE**

Select the alternative A, B, C or D that best answers the question and indicate your choice by shading the appropriate letter.

|    |                         |                         |                         |                         |
|----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 2. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 3. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 4. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 5. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 6. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |

**Multiple Choice – Select the most correct answer**

1. Solve for  $x$ :  $2(3x - 2) = x + 4(x - 1)$

(A)  $-\frac{8}{11}$

(B) 0

(C) 8

(D) All real  $x$

2. Select the appropriate values of  $x$  for the inequation:

$$(1 - x)(x - 4) < 0$$

(A)  $x < 1$

(B)  $x > -1$

(C)  $1 < x < 4$

(D)  $x < 1$  or  $x > 4$

3. Which of the following trigonometric graphs does not have  $x$  – intercepts?

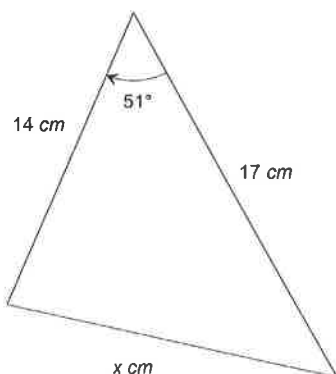
(A)  $y = \sin x$

(B)  $y = \tan x$

(C)  $y = \cot x$

(D)  $y = \sec x$

4. What is the length of  $x$  in the diagram below?



(A)  $\sqrt{14^2 + 17^2 + 476 \cos 51^\circ}$

(B)  $\sqrt{14^2 + 17^2 - 476 \cos 51^\circ}$

(C)  $\sqrt{14^2 + 17^2 - 476 \sin 51^\circ}$

(D)  $\sqrt{14^2 + 17^2 + 476 \sin 51^\circ}$

5. How many solutions are there to the equation  $\sin x = 0$  for  $0 \leq x \leq 360^\circ$

(A) 0

(B) 1

(C) 2

(D) 3

6. Solve for  $x$ :  $\sin(x - 60^\circ) = -\frac{1}{2}$ , for  $0^\circ \leq x \leq 180^\circ$

(A)  $30^\circ$

(B)  $60^\circ$

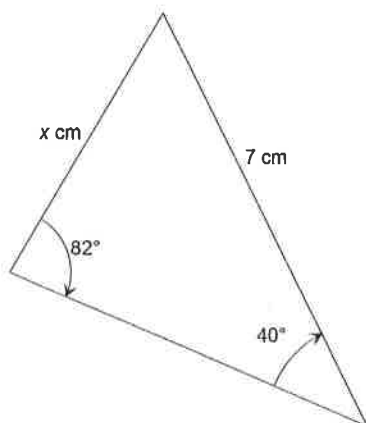
(C)  $90^\circ$

(D)  $210^\circ$

Extended Response – Show full working out

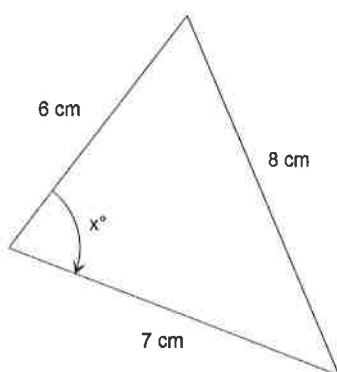
Question 7

- (a) Find the value of  $x$  to 1 decimal place



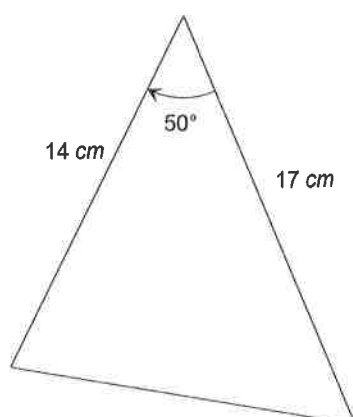
2

- (b) Find the value of  $x$  to the nearest degree



2

- (c) Find the area of the triangle below (answer to 1 decimal place).



2

- (d) In  $\triangle ABC$ ,  $AC = 30$  cm,  $BC = 20$  cm and  $\angle BAC = 40^\circ$ . Find the value(s) of  $\angle ABC$  to the nearest minute.

3

### Question 8

(a) Find the exact values of (express your answer in simplest form):

(i)  $\cos 150^\circ$  1

(ii)  $\frac{\sec 45^\circ + \cos 30^\circ}{\operatorname{cosec} 210^\circ}$  2

(b) Find  $\theta$  to the nearest minute for  $0^\circ \leq \theta \leq 360^\circ$ :

(i)  $\sin \theta = -0.234$  1

(ii)  $\sin \theta = 4 \cos \theta$  2

(iii)  $4 \cos 2\theta = 2$  3

### Question 9

(a) State the period and amplitude and sketch

$y = 2 \cos 3x$ , for  $0^\circ \leq x \leq 360^\circ$  3

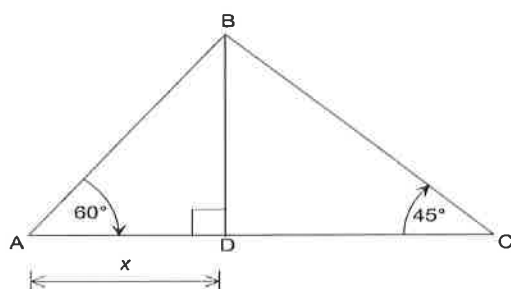
showing all relevant features.

(b) Solve for  $x$ :

$\tan(2x + 20^\circ) = \cot(3x - 60^\circ)$  2

(c) Given that  $\tan \theta = 3.5$  and  $\cos \theta < 0$ , find the exact value of  $\sin \theta$  2

(d) In the diagram below  $AD = x$ . Find the length of  $BC$  in terms of  $x$ .



2

### Question 10

(a) Find the points of intersection of:  $y = (x - 2)^2$  and  $y = 3x - 6$ . 3

(b) Solve for  $x$  and sketch your solution on a number line:

$\frac{5}{2-x} \leq 3$  3

(c) Solve for  $x$ :

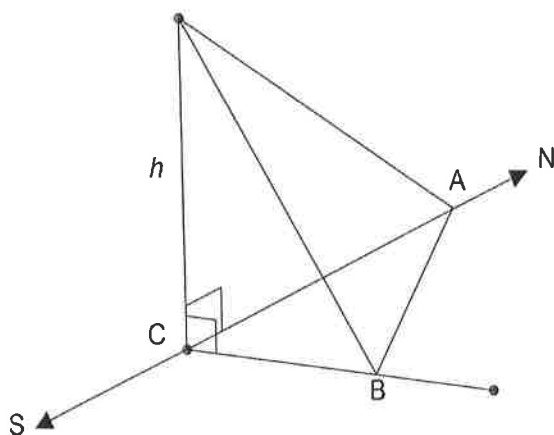
$|x + 1| + |x - 2| = 5$  3

### Question 11

- (a) A is 5 km N37°W from a lighthouse and B is 12 km S53°W from the same lighthouse.
- (i) Find the distance from A to B. 2
- (ii) Find the bearing of B from A. 2
- (b) Simplify:
- $$1 - \sin \theta \cos(90^\circ - \theta) \quad \text{2}$$
- (c) Prove the identity:
- $$(\cot \theta + \operatorname{cosec} \theta)^2 = \frac{1 + \cos \theta}{1 - \cos \theta} \quad \text{3}$$

### Question 12

- (a) Solve for  $\theta$  where  $0^\circ \leq \theta \leq 360^\circ$
- $$2 \tan^2 \theta = 5 \sec \theta - 4 \quad \text{3}$$
- (b) Amy walks away from a telegraph pole in a northerly direction. When she arrives at Point A the angle of elevation to the top of the pole is  $23^\circ$ . Brenda walks in a bearing of  $032^\circ T$  from the same tower until she reaches Point B and notices that the angle of elevation to the top of the pole is  $17^\circ$ . The distance between A and B is 55m. Let  $h$  be the height of the pole and Point C be the base of the pole.



- (i) Copy the diagram above into your answer booklet and clearly mark all of the information given. 1
- (ii) Find expressions for  $AC$  and  $BC$  in terms of  $h$  2
- (iii) Hence, or otherwise, find the height  $h$  of the tower to the nearest metre 3

## Multiple Choice

1. B

2. D

3. D

4. B

5. D

6. A

# Question 7

$$(a) \frac{x}{\sin 40^\circ} = \frac{7}{\sin 82^\circ}$$

✓ correct sine rule

$$\therefore x = \frac{7 \sin 40^\circ}{\sin 82^\circ}$$

$$\therefore x = 4.5 \text{ cm (1 d.p.)}$$

✓ answer

$$(b) \cos x^\circ = \frac{6^2 + 7^2 - 8^2}{2 \times 6 \times 7}$$

✓ correct cosine rule

$$x^\circ = \cos^{-1}\left(\frac{1}{4}\right)$$

$$\therefore x^\circ = 76^\circ \text{ (nearest degree)}$$

✓ correct answer

$$(c) A = \frac{1}{2} \times 14 \times 17 \times \sin 50^\circ$$

$$= 91.2 \text{ cm}^2 \text{ (1 d.p.)}$$

✓ correct formula

✓ correct answer

$$(d) \frac{\sin B}{30} = \frac{\sin 40^\circ}{20}$$

✓ correct substitution into formula

$$\therefore \sin B = \frac{30 \sin 40^\circ}{20}$$

$$\therefore B = 74^\circ 37' \text{ (nearest minute)}$$

✓ acute answer

$$B = \text{or } 105^\circ 23' \text{ (nearest minute)}$$

✓ obtuse answer

# Question 8

(a)(i)  $\cos 150^\circ = -\sqrt{3}/2$

✓ correct answer

(a)(ii)  $\frac{\sqrt{2} + \sqrt{3}/2}{(-2)}$

✓ at least 2 correct exact values

$= \frac{-2\sqrt{2} - \sqrt{3}}{4}$

✓ correct answer

(b)(i)  $\theta = 193^\circ 32'$

or  
 $\theta = 346^\circ 28'$

✓ both answers

(b)(ii)  $\sin \theta = 4 \cos \theta$

$\therefore \tan \theta = 4$

✓  $\tan \theta = 4$

$\therefore \theta = 75^\circ 58'$

or  
 $\theta = 255^\circ 58'$

✓ both answers

(b)(iii)  $4 \cos 2\theta = 2$

$\therefore \cos 2\theta = 1/2$

$\therefore 2\theta = 60^\circ, 300^\circ, 420^\circ, 660^\circ$

$\therefore \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

✓ 2 answers for  $2\theta$

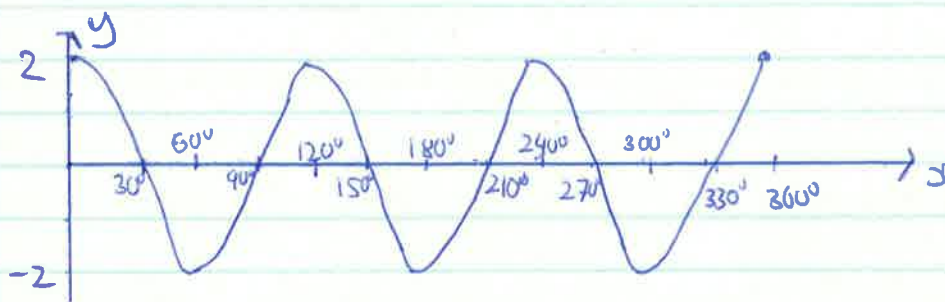
✓ 4 answers for  $2\theta$

✓ all correct answers

## Question 9

(a) period =  $120^\circ$

amplitude = 2



✓ period

✓ amplitude

✓ graph

(b)  $\tan(2x + 20^\circ) = \cot(3x - 60^\circ)$

$\therefore 2x + 20^\circ = 90 - (3x - 60^\circ)$

$2x + 20^\circ = 90 - 3x + 60^\circ$

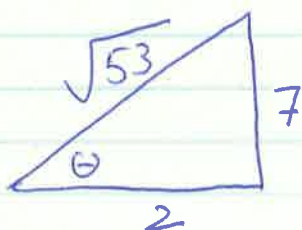
$5x = 130^\circ$

$x = 26^\circ$

✓ complementary angles

✓ correct answer

(c)  $\tan \theta = 7/2$



$\tan \theta > 0$ ,  $\cos < 0 \rightarrow$  3rd quadrant

$\sin \theta = -\frac{7}{\sqrt{53}}$

✓ correct value

✓ correct sign

$$9.(d) \tan 60^\circ = BD/x$$

$$BD = x \tan 60^\circ$$

$$BD = \sqrt{3} x$$

✓ length BD

$$\sin 45^\circ = BD/BC$$

$$BC = \frac{BD}{(1/\sqrt{2})}$$

$$BC = (\sqrt{3} x)(\sqrt{2})$$

$$BC = \sqrt{6} x$$

✓ correct answer

## Question 10

(a)  $y = (x-2)^2$  &  $y = 3x-6$

$$(x-2)^2 = 3x-6$$

$$x^2 - 4x + 4 = 3x - 6$$

$$x^2 - 7x + 10 = 0$$

$$(x-5)(x-2) = 0$$

$$x = 5 \text{ or } x = 2$$

$$y = 3 \times 5 - 6 \text{ or } y = 3 \times 2 - 6$$
$$= 9 \qquad \qquad \qquad = 0$$

∴ intersection at  $(5, 9)$  &  $(2, 0)$

✓ simultaneous equation

✓ correct x-values

✓ correct points

(b)  $\frac{5}{2-x} \leq 3$

$$\frac{5(2-x)^2}{\cancel{(2-x)}} \leq 3(2-x)^2, \quad x \neq 2$$

$$5(2-x) \leq 3(2-x)^2$$

$$5(2-x) - 3(2-x)^2 \leq 0$$

$$(2-x)(5 - 3(2-x)) \leq 0$$

$$(2-x)(5-6+3x) \leq 0$$

$$(2-x)(3x-1) \leq 0$$

$$\therefore x \leq \frac{1}{3} \text{ or } x > 2$$

✓ any valid method

✓ points  $\frac{1}{3}$  & 2

✓ correct inequalities

(c) case 1  $x \leq -1$

$$-(x+1) - (x+2) = 5$$

$$-x - 1 - x + 2 = 5$$

$-1-2$

$$-2x + 1 = 5$$

$$-2x = 4$$

$$\therefore x = -2$$

✓ any  
correct  
method

$$\checkmark x = -2$$

case 2  $-1 \leq x \leq 2$

$$(x+1) - (x-2) = 5$$

$$x+1 - x + 2 = 5$$

$$3 = 5$$

$\therefore$  no solutions

case 3  $x \geq 2$

$$(x+1) + (x-2) = 5$$

$$\therefore 2x - 1 = 5$$

$$2x = 6$$

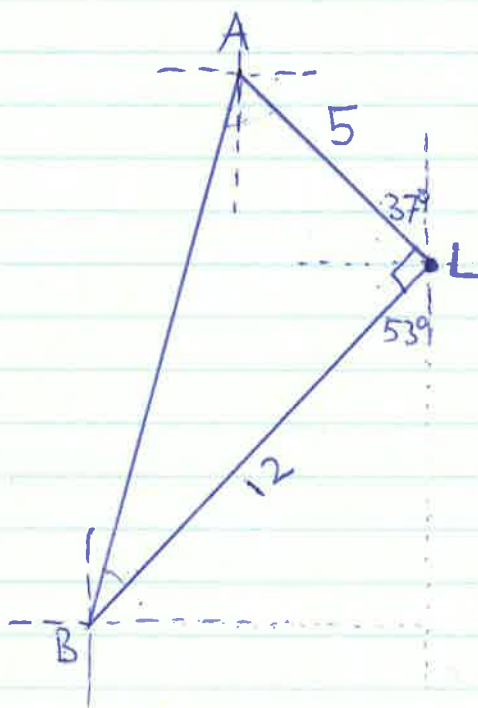
$$\therefore x = 3$$

$$\checkmark x = 3$$

$\therefore$  solutions are  $x = -2$  or  $x = 3$

# Question 11

(a)(i)



$$AB = \sqrt{5^2 + 12^2}$$

$$= 13 \text{ km}$$

✓ pythagoras

✓ answer.

(ii)  $\angle BAL = \theta$

$$\tan \theta = \frac{12}{5}$$

$$\therefore \theta = 67^\circ 23' \text{ (nearest minute)}$$

✓ finding  $\angle A$

$$\text{Bearing} = 180^\circ + 67^\circ 23' - 37^\circ$$

$$= 210^\circ 23' \text{ T (nearest minute)}$$

✓ correct answer

$$(b) 1 - \sin \theta \cos (90 - \theta)$$

$$= 1 - \sin^2 \theta$$

$$= \cos^2 \theta$$

✓  $\cos(90 - \theta) = \sin \theta$

✓ correct answer

$$(c) (\cot \theta + \operatorname{cosec} \theta)^2$$

$$= \cot^2 \theta + 2 \cot \theta \operatorname{cosec} \theta + \operatorname{cosec}^2 \theta$$

$$= \frac{\cos^2 \theta}{\sin^2 \theta} + 2 \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} + \frac{1}{\sin^2 \theta}$$

✓ correctly converting ratios

$$= \frac{\cos^2 \theta + 2 \cos \theta + 1}{\sin^2 \theta}$$

$$= \frac{(\cos \theta + 1)^2}{1 - \cos^2 \theta}$$

✓ everything in terms of  $\cos \theta$

$$= \frac{(\cancel{\cos \theta} + 1)(\cos \theta + 1)}{(1 + \cancel{\cos \theta})(1 - \cos \theta)}$$

$$= \frac{1 + \cos \theta}{1 - \cos \theta}, \text{ as required}$$

✓ final answer

## Question 12

(a)  $2\tan^2 \theta = 5\sec \theta - 4$

$$2(\sec^2 \theta - 1) = 5\sec \theta - 4$$

$$2\sec^2\theta - 2 = 5\sec\theta - 4$$

$$2\sec^2 \theta - 5\sec \theta + 2 = 0$$

$$(\sec \theta - 2)(2 \sec \theta - 1) = 0$$

$$\sec \theta = 2 \text{ or } \sec \theta = \frac{1}{2}$$

as  $|\sec \theta| \geq 1$

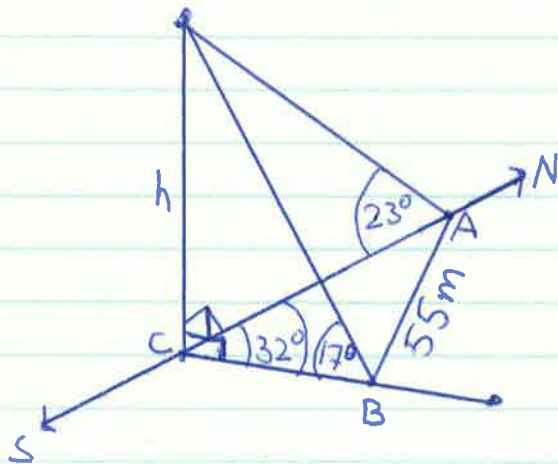
$$\therefore \theta = 60^\circ \text{ or } \theta = 300^\circ$$

✓ quadratic equation  
in sec  $\theta$

✓ values for secG

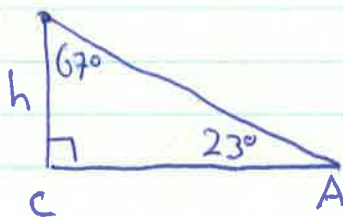
✓ values  
for  $G$

(b)(i)



✓ all information shown

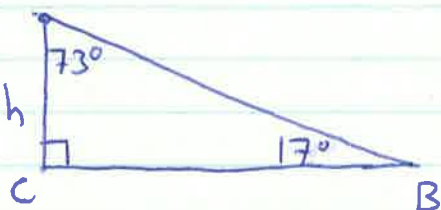
(b)(i)



$$\tan 67^\circ = \frac{AC}{h}$$

$$\therefore AC = h \tan 67^\circ$$

✓ AC

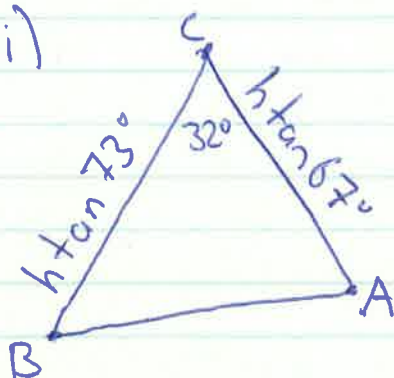


$$\tan 73^\circ = \frac{BC}{h}$$

$$\therefore BC = h \tan 73^\circ$$

✓ BC

(b)(iii)



$$55^2 = h^2 \tan^2 73^\circ + h^2 \tan^2 67^\circ - 2(h \tan 73^\circ)(h \tan 67^\circ) \cos 32^\circ$$

✓ correct  
cosine  
rule

$$55^2 = h^2 (\tan^2 73^\circ + \tan^2 67^\circ - 2 \tan 73^\circ \tan 67^\circ \cos 32^\circ)$$

$$\therefore h^2 = \frac{55^2}{\tan^2 73^\circ + \tan^2 67^\circ - 2 \tan 73^\circ \tan 67^\circ \cos 32^\circ}$$

✓ making  
 $h^2$  the  
subject

$$\therefore h = 31 \text{ m (nearest m)}$$

✓ final  
answer